



United International University

Department of Computer Science and Engineering

CSE 4495: Software Testing and Quality Assurance

Final Examination : Spring 2025

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. Numbers to the right of the questions denote their marks.

1. (a) A major bank has launched a new mobile banking application. During the first week, the following issues are reported: [4]
- On payday, when user traffic is highest, the app becomes extremely slow, and many users experience timeouts and connection errors.
 - Users report that transferring money between accounts sometimes results in the amount being deducted from the source account but not credited to the destination account for several hours.
 - A security researcher discovers a flaw where, by manipulating the network request, a user can view the transaction history of a different user.

As a Quality Assurance manager, identify three critical quality attributes that have been compromised in this scenario. For each attribute, name it and explain how the scenario demonstrates its failure.

- (b) As a QA lead, you are tasked to evaluate the reliability of two different systems.
- System X:** An anti-lock braking system (ABS) for a new car model. In a controlled test environment, the ABS was activated 8,000 times under various conditions. It failed to engage correctly on 5 of those activations.
 - System Y:** A real-time financial data feed that provides stock market updates to clients. It operates continuously. Over a 5-day period (120 hours), the system crashed and had to be restarted 18 times.

For each systems (X and Y), you must do the following:

- Identify the most appropriate reliability metric to use from the following choices: *POFOD* (Probability of Failure on Demand) or *ROCOF* (Rate of Occurrence of Failure). Justify your choice based on the nature and usage pattern of the system. [2]
 - Calculate the value of the chosen metric. [2]
- (c) Discuss the difference between Verification and Validation in the context of software testing. For each term, provide a clear and brief definition and an example of a corresponding testing activity. [2]
2. You are tasked with testing a new “Smart Access” security system for an office building. The system controls access to different offices based on an employee’s role, the time of day, and the current security level of the building. The logic for granting or denying access at a specific door follows these requirements:

R1: Access is always denied if the building’s security level is set to ‘Lockdown’, regardless of any other conditions. This rule has the highest priority.

R2: All employees, regardless of role, are granted access to the ‘Lobby’ zone during standard business hours (9 AM - 5 PM). Outside these hours, they need specific authorization.

R3: Employees with the role ‘Engineer’ or ‘Executive’ are granted 24/7 access to the ‘General Office’ zone. Employees with the role ‘Staff’ are only granted access during standard business hours.

R4: Access to the ‘Server Room’ zone is restricted to ‘Engineers’ only, but an additional condition applies: their request must be accompanied by a valid, one-time access token. Without the token, even Engineers are denied.

R5: If the building’s security level is ‘High Alert’, access is only granted to ‘Executives’, and only for the ‘General Office’ and ‘Lobby’ zones. All other requests are denied.

The system will respond to an access request with one of the following outputs:

- Access Granted:** The door is unlocked.
- Access Denied:** The door remains locked.
- Error: Invalid Input:** The data is invalid (e.g., the employee ID is not in the system, the zone does not exist).

- (a) Identify your choices while testing this feature. Choices are aspects that you have control over that can alter the output of the program. Also, enumerate representative values for your choices. [6]
- (b) Using your choices and values, first calculate the total number of test specifications required for an exhaustive test. Then, apply the Category Partition Method (CPM) constraints ([ERROR], [IF], [SINGLE]) based on the requirements to find the reduced number of test specifications. You must state the constraints you are applying and which choices/values they affect. [6]
3. You are writing unit tests for a new Library Management System. The core logic involves a `LibraryMember` checking out a `Book`. The system has specific rules: a member cannot borrow more books than their limit, and they cannot borrow a book that is already checked out by someone else. The `borrowBook(Book book)` method contains the business logic that you need to test. It checks the member's limit and the book's availability before completing the transaction. This method modifies the state of both the `LibraryMember` and the `Book` objects.

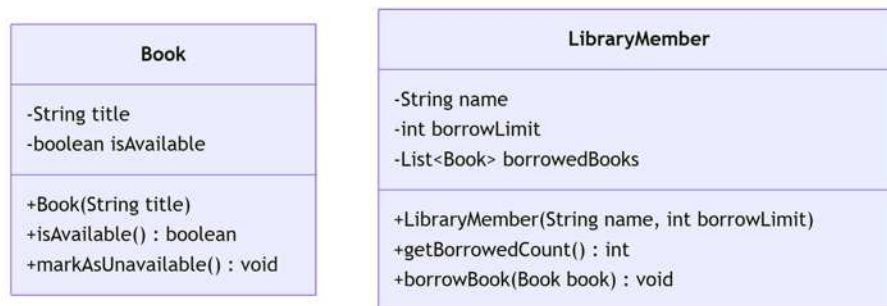


Figure 1: Class diagrams for Ques-3

- (a) Write a complete JUnit test to verify the successful scenario where a library member borrows a book. The test must create a `LibraryMember` and an available `Book`. After calling the `borrowBook` method, it must assert that: The member's count of borrowed books has increased correctly and the book's status has correctly been marked as unavailable. [4]
- (b) Write a test case named verifies that a member cannot borrow a book if it has already been borrowed by another member. Invoking the `LibraryMember.borrowBook(Book book)` method in this case a `BorrowingExceededException` with the exact message "The book is currently unavailable". [4]
4. Consider the following code for a `calculateShippingFee` method that calculates the delivery fee for an e-commerce platform:

```

1. public double calculateShippingFee(double orderValue, boolean isPremiumMember) {
2.     double fee = 10.0; // Default shipping fee
3.
4.     if (orderValue > 100.0 || isPremiumMember) {
5.         fee = 5.0; // Reduced fee for eligible customers
6.     }
7.
8.     if (orderValue > 200.0) {
9.         fee = 0.0; // Free shipping
10.    }
11.    return fee;
12. }
  
```

- (a) Draw the Control Flow Graph (CFG) for the `calculateShippingFee` method. [4]
- (b) Consider the two test cases :
- TC1 - `orderValue = 150.0`, `isPremiumMember = false`
 - TC2 - `orderValue = 250.0`, `isPremiumMember = true`

Trace the execution path for both TC1 and TC2 in your CFG. After executing each test, calculate the branch and condition coverage of the test suite. if complete coverage for both criteria is not achieved, design one new test case to achieve full coverage. Justify why your new test case achieves this goal. [3+3=6]